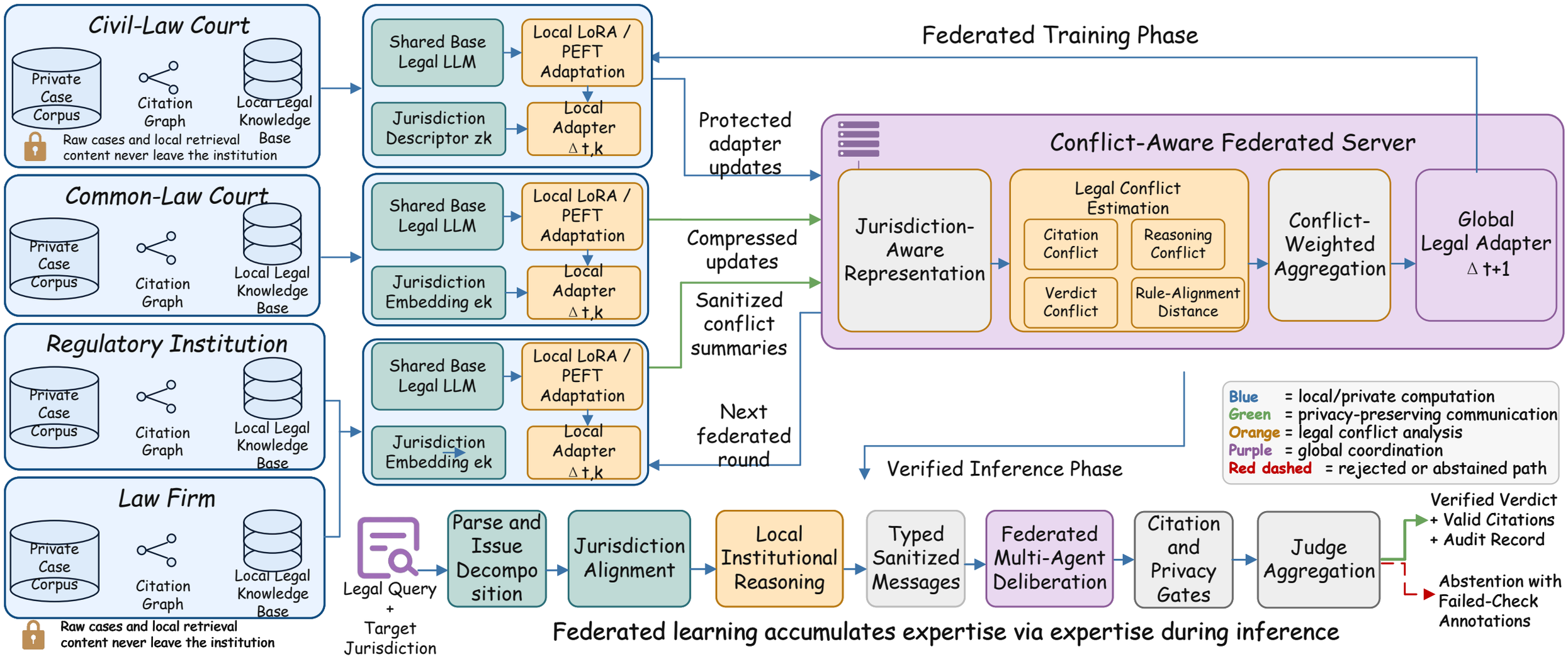

Civil-Law Court
Private Case Corpus
Citation Graph
Local Legal Knowledge Base
Raw cases and local retrieval content never leave the institution
Common-Law Court
Private Case Corpus
Citation Graph
Local Legal Knowledge Base
Regulatory Institution
Private Case Corpus
Citation Graph
Local Legal Knowledge Base
Law Firm
Private Case Corpus
Citation Graph
Local Legal Knowledge Base
Raw cases and local retrieval content never leave the institution
Shared Base Legal LLM
Local LoRA / PEFT Adaptation
Jurisdiction Descriptor zk
Local Adapter Δ t,k
Shared Base Legal LLM
Local LoRA / PEFT Adaptation
Jurisdiction Embedding ek
Local Adapter Δ t,k
Shared Base Legal LLM
Local LoRA / PEFT Adaptation
Jurisdiction Embedding ek
Local Adapter Δ t,k
Federated Training Phase
Protected adapter updates
Compressed updates
Sanitized conflict summaries
Next federated round
Conflict-Aware Federated Server
Jurisdiction-Aware Representation
Legal Conflict Estimation
Citation Conflict
Reasoning Conflict
Verdict Conflict
Rule-Alignment Distance
Conflict-Weighted Aggregation
Global Legal Adapter Δ t+1
Verified Inference Phase
Blue = local/private computation
Green = privacy-preserving communication
Orange = legal conflict analysis
Purple = global coordination
Red dashed = rejected or abstained path
Legal Query + Target Jurisdiction
Parse and Issue Decomposition
Jurisdiction Alignment
Local Institutional Reasoning
Typed Sanitized Messages
Federated Multi-Agent Deliberation
Citation and Privacy Gates
Judge Aggregation
Verified Verdict + Valid Citations + Audit Record
Abstention with Failed-Check Annotations
Federated learning accumulates expertise via expertise during inference